Utah Division of Water Rights

Technology Modernization Roadmap



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Disclaimer

This report was prepared for the Utah Division of Water Rights (DWRi) by Utah State University. The roadmap presented in this report is based on a Hydroinformatics and Technology Gap Analysis completed by USU for DWRi and presents a potential pathway for DWRi to address the gaps that were identified and modernize their data collection and management infrastructure. While this work was completed in collaboration with DWRi, any opinions, findings, conclusions, or recommendations in this report are those of the authors, are submitted for consideration by DWRi, and do not necessarily reflect the views of DWRi.

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Executive Summary

The Utah Division of Water Rights (DWRi) is an agency of Utah State Government within the Department of Natural Resources that administers the measurement, appropriation, apportionment, and distribution of the state's valuable water resources. DWRi collects and maintains records of water diversion from surface and underground sources, which are mainly included in water distribution systems within the state. There are 44 active river distribution systems across the state, each of which is managed by a River Commissioner and some of which have automated systems to measure water diversion/use and or operate controls.

The scope of DWRi's measurement data includes thousands of diversion locations at which measurements of flow are recorded, along with a Water Use Program dataset that details water use by public and industrial water suppliers in the state, among other ongoing data collection. In addition to these current operations, DWRi has been tasked with new data collection and management activities associated with Utah's Agricultural Water Optimization Program and will likely be adding significant new data collection and management associated with recommendations from the recent Legislative Audit (Utah Office of the Legislative Auditor General, 2023a) and ongoing data gap analyses within the state (e.g., Lukens et al., 2024). While DWRi manages several other important datasets, this Technology Modernization Roadmap and the Gap Analysis on which it is based (Horsburgh et al., 2024) focused on measurement data (e.g., observations of flow recorded at stream diversions), Water Use Program data (water diversion and use data primarily from public water suppliers), and water distribution accounting (the procedures used by DWRi to ensure that water is distributed to water users by priority of legal water rights).

The extensive data collection DWRi undertakes to enable management of Utah's water resources and to meet its mandate requires significant information technology (IT) infrastructure. Data collection hardware (e.g., measurement structures and sensors installed in the field), software, and databases are needed to enable collection, management, and use of the data. While DWRi has developed and operates a water use data infrastructure for the state of Utah, some of the systems used by DWRi have become dated, lack necessary flexibility, and may not scale to meet DWRi's growing data management needs. Furthermore, the recent Legislative Audit (Utah Office of the Legislative Auditor General, 2023a) recommended that DWRi work toward use of best practices for data management to improve effectiveness, increase the availability and usability of DWRi's data, and to more effectively monitor state water. To address these challenges and recommendations, in 2023 DWRi initiated a process to critically examine their existing data infrastructure and identify areas where modernization and

enhancements are needed. That initial effort resulted in a "Hydroinformatics[1] and Technology Gap Analysis" conducted by Utah State University that provided an in-depth examination of DWRi's current data and IT infrastructure, software systems, and processes (Horsburgh et al., 2024). The Gap Analysis identified disparities between DWRi's existing systems and capabilities and the organizational goals, industry standards, and improvements needed as identified by USU and DWRi personnel.

This Technology Modernization Roadmap considers the gaps and recommendations developed through the Gap Analysis and lays out a sequenced plan, or roadmap, for software development and other activities that will help DWRi modernize and advance their data collection and management infrastructure to meet existing operational needs, along with evolving data collection and management requirements associated with new programs. This Roadmap lays out the foundations for implementing data management best practices to improve the usability of DWRi's data, along with the transparency and accountability of DWRi's operations for effective water management and planning in Utah for decades to come.

[1] Hydroinformatics is the study, design, development, and deployment of hardware and software systems for hydrologic data collection, distribution, interpretation, and analysis to aid in the understanding and management of water in the natural and built environment.



1. Introduction

Data collection is necessary for effectively managing water resources. In Utah, the state's Division of Water Rights (DWRi) operates data infrastructure aimed at collecting, retrieving, and managing water use data from monitoring sites throughout the state. The scope of DWRi's measurement data includes approximately 1,550 active surface water diversion stations with approximately 500 of those having automated data telemetry and data loggers. These monitoring sites are operated by DWRi or by many partner organizations to provide Utah's water use data infrastructure, and the data produced serve as the basis on which water rights and water distribution accounting decisions are made on a daily basis. DWRi cooperates with the United States Geological Survey (USGS) to operate 79 stream gages (out of the 1,550 stations) in various parts of the state. DWRi also collects surface water use data from 908 sources and groundwater use data from 1,740 wells. DWRi also maintains water use records on about 1,400 public, secondary, industrial water suppliers, and other water users who self-report their diversion data.

In operating the state's water use data infrastructure, DWRi faces several challenges that fall within the Hydroinformatics area, including installing and operating a large and growing number of measurement sites, integrating inconsistent data from many different sources, ensuring the quality of measurement data, data analysis and visualization, operating and maintaining a modern cyberinfrastructure for data collection and management, delivering data to distribution accounting models in near real time, interfacing with partners who want to use data managed by DWRi, and others.

Given the importance of the data collection and management infrastructure operated by DWRi, the agency initiated a Hydroinformatics and Technology Gap Analysis in 2023 to develop an assessment of the current state of DWRi's data and information technology (IT) infrastructure, software, systems, and processes, with the objective of identifying critical gaps between existing capabilities and DWRi's organizational goals, industry standards, and needed improvements identified by DWRi personnel (Horsburgh et al., 2024). The Gap Analysis, which was performed by the Utah Water Research Laboratory at Utah State University, focused not only on data collection and management, but also on the software, database systems, and software development processes used by DWRi to develop, operate, and maintain the cyberinfrastructure required to track and account for water use within the state of Utah. The Gap Analysis also addressed elements of and recommendations from the recent Legislative Audit addressing data integrity, program best practices, and comprehensive water planning (Utah Office of the Legislative Auditor General, 2023a) and the Utah Office of the Legislative Auditor General's "High Risk List" of critical vulnerabilities in Utah, which identified Utah's water use and water infrastructure needs as two of twelve high-risk areas in Utah (Utah Office of the Legislative Auditor General, 2023b).

The comprehensive view of DWRi's operations provided by the Hydroinformatics Gap Analysis was necessary given the scale of the data collected and managed by DWRi and the age and complexity of the IT systems that support DWRi's operations. It was also necessary to provide a comprehensive picture of the current state of DWRi's systems such that DWRi can modernize and plan for future needs consistent with the vision and actions laid out in DWRi's 2024–2025 Strategic Action Plan (Utah Division of Water Rights, 2024). While DWRi manages several other important datasets, the Gap Analysis focused on measurement data (e.g., observations of flow recorded at stream diversions), Water Use Program data (water diversion and use data primarily from public water suppliers), and water distribution accounting (the procedures used by DWRi to ensure that water is distributed to water users by priority of legal water rights).

The Gap Analysis resulted in identification of 41 specific gaps and 66 specific recommendations developed to address those gaps. However, while the Gaps and Recommendations in the Gap Analysis are informative, the Gap Analysis does not provide a specific prioritization of those gaps and recommendations or a sequenced set of activities that DWRi can perform to address them. Thus, the purpose of this document is to assess the gaps and recommendations in the Gap Analysis and organize them into an actionable set of tasks and activities that DWRi can take on to modernize and advance their cyberinfrastructure and systems for data collection, management, sharing, and internal use. This roadmap is intended to assist DWRi in efficiently allocating resources to meet priority needs. It should be noted that this Technology Modernization Roadmap addresses all of the gaps from the Gap Analysis, at least in part, but may not fully address all of the gaps and recommendations described in the Gap Analysis document. Thus, DWRi may later revisit the Gap Analysis and identify additional tasks or actions required based on the gaps and recommendations documented there.



2. Methods

To develop this roadmap, we examined the gaps and recommendations identified in our Gap Analysis document. Each identified gap has one or more stated recommendations for how DWRi can address that gap. USU's team individually ranked each recommendation from the gap analysis against a set of criteria that were ideally chosen to help us identify the most important gap/recommendation combinations. Criteria used for evaluating each recommendation included:

- Degree to which technical debt[2] would be reduced
- Impact on data integrity and accuracy
- Impact on data accessibility and availability
- Impact on software system reliability and performance
- Alignment with DWRi's priorities and 2024–2025 Action Plan
- Potential risk if the recommendation was left unaddressed
- Impact on DWRi's operations
- Impact on long-term sustainability and scalability of DWRi's software systems

Composite scores for each recommendation were calculated, and then an overall composite score was calculated for each recommendation by averaging the composite scores from each team member at USU. While these composite scores provide some idea of the relative importance of each recommendation, many of the recommendations received high composite scores, indicating our team's perceptions that many of the recommendations were of high importance. The relative similarity in composite scores across many of the recommendations diminished their utility alone in prioritizing individual recommendations. Furthermore, some of the recommendations from the Gap Analysis are interdependent, which means that they would need to be sequenced in a particular order for them to be addressed/implemented (e.g., it is possible that a recommendation ranked as highly important based on its composite score might only be possible after a different recommendation that had a lower composite score is

[2] Technical debt is a concept in software development that refers to the cost an organization incurs when it makes decisions or compromises in the short term to expedite the development or release of a software product (e.g., to add a new feature or meet a new requirement) without addressing longer term architectural or development needs. These short term or isolated decisions typically involve writing code that may be less efficient or less maintainable. Failing to address technical debt can lead to increased maintenance costs, slower development cycles, reduced software quality, and increased risk of software failures. Just as financial debt accrues interest over time, technical debt accrues "interest" in the form of increased effort, complexity, and costs in the future.

implemented). Thus, it was not clear cut to simply prioritize recommendations based on their numeric composite scores.

To enhance the prioritization, we performed a qualitative analysis of the recommendations from the Gap Analysis to group them into logical areas, where the logical groups of recommendations could then be mapped to a major task or set of tasks that would become the basis for this Technology Modernization Roadmap. The major tasks and their associated sub-tasks laid out in this document represent a sequenced and strategic plan outlining a process DWRi could use for updating, enhancing, and modernizing their existing cyberinfrastructure. Major tasks and subtasks were designed to address as many of the gaps and recommendations from the Gap Analysis as possible. This roadmap is designed such that completion of the major task and its associated subtasks will address all of the recommendations linked to that task. In the description of the tasks and subtasks that follow, we have indicated the specific gaps and recommendations from the Gap Analysis that each task or subtask addresses. Readers should consult the Gap Analysis document (Horsburgh et al., 2024) for specific details of gaps and recommendations.

3. Technology Modernization Roadmap

Figure 1 shows a sequenced overview of the Technology Modernization roadmap based on our analysis of the gaps and recommendations in the Gap Analysis. Each of the numbered sections depicted in Figure 1, along with the item related to stakeholder engagement, are considered to be major “task groups” under which specific tasks and subtasks are specified in the sections that follow. The numbered sequence of the task groups suggests an order of operations that is based on both suggested priority and interdependence of tasks—e.g., some tasks must be finished before others can begin.

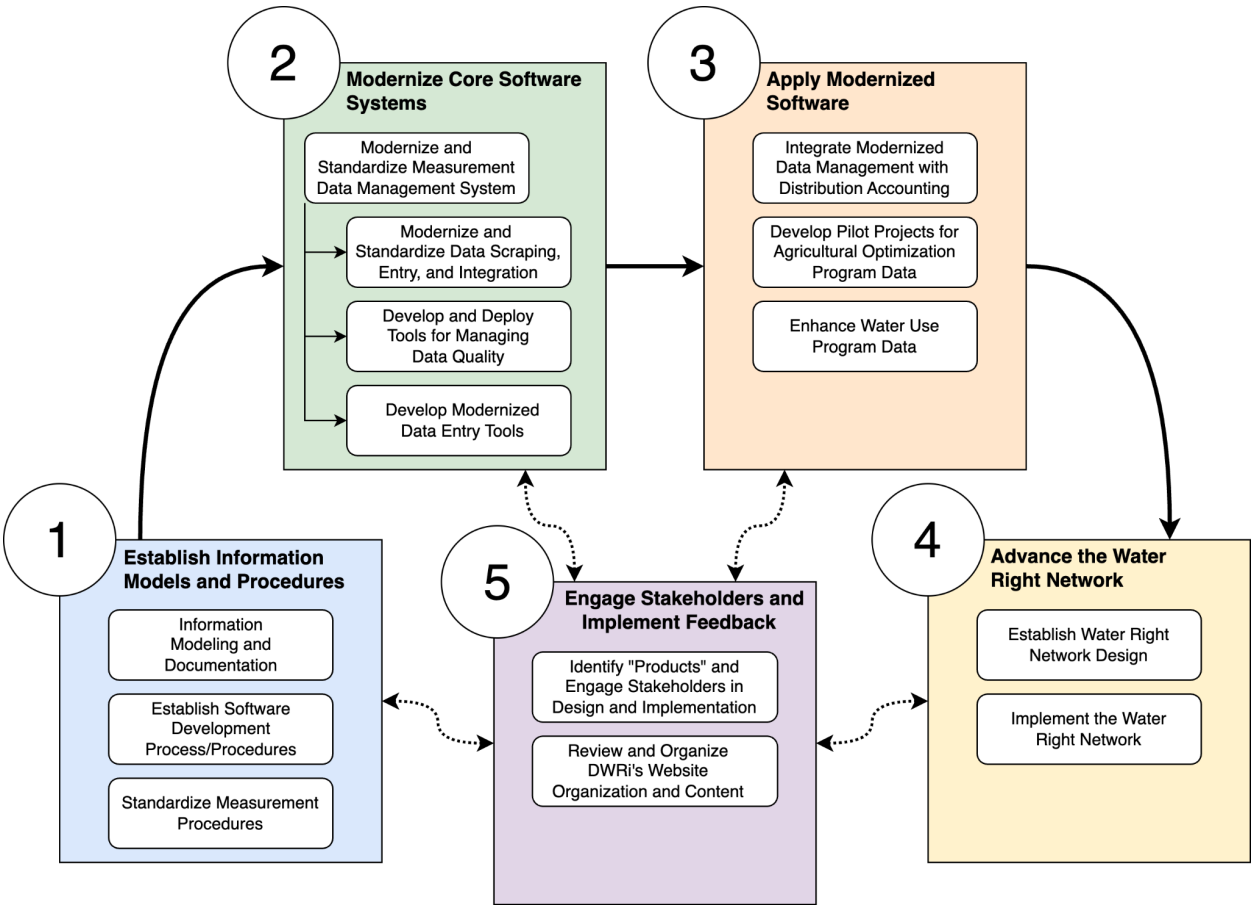


Figure 1. Overview of technology modernization roadmap.

TASK GROUP 1: ESTABLISH INFORMATION MODELS AND PROCEDURES

The Gap Analysis pointed out that DWRi lacks formal documentation of the information and data models used to represent the data they use operationally, which may lead to uncertainty in some instances where metadata are incomplete. Thus, one of the most important first steps in this Technology Modernization Roadmap is to formalize, document, and, where necessary, modernize the information and data models used by DWRi for storing and exchanging data. While these information models exist to some degree in the various database structures, file formats, and other data encodings already in use by DWRi, this major task group is an opportunity for DWRi to formalize them in a way that promotes shared understanding among current and potential future employees and ensures consistency and completeness in data and metadata wherever data are used—e.g., data input from partners, data stored in operational databases, data used in distribution accounting, data downloaded by interested stakeholders, etc. This activity is prioritized up front so that documentation of DWRi's information and data models can serve as blueprints for subsequent software development/modernization efforts.

Another major task that we prioritize up front is for DWRi to more formally establish their software development, testing, and deployment procedures. The Gap Analysis identified opportunities for DWRi to better take advantage of modern software development tools. Establishing these procedures early in DWRi's data system modernization efforts is important to ensure that subsequent software development activities adhere to these procedures.

Developing standards and specifications for measurement procedures is an important task for improving the quality of data collected by and submitted to DWRi. We prioritize this task within this first task group because this task can be completed independent of other software development and data system modernization efforts and because completing this task as early as possible will establish early benefit by improving the overall quality of measurement data assembled by DWRi. Each of the major tasks prioritized in this task group, along with recommended subtasks, are summarized in Figure 2 and described in the following sections.

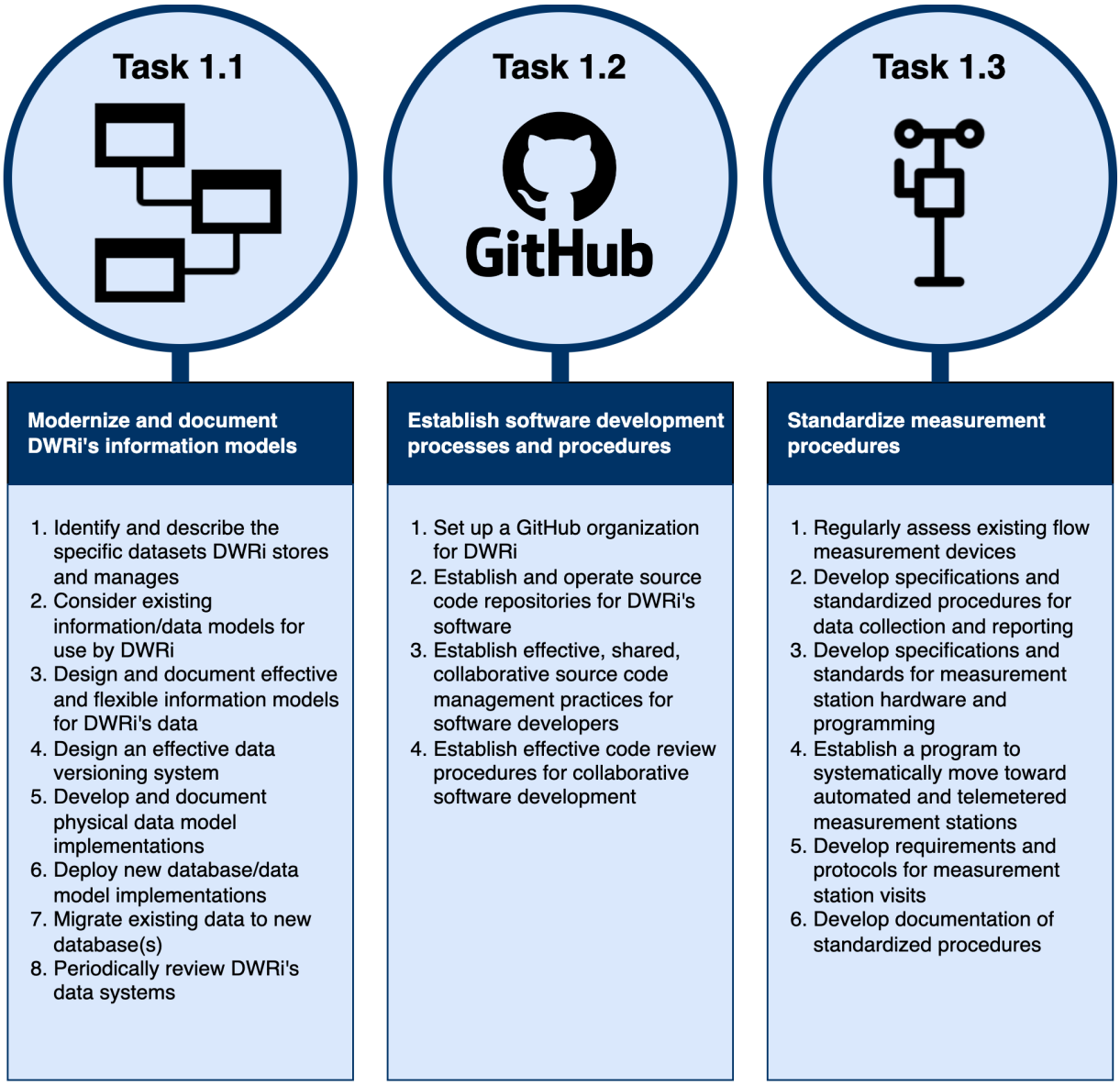


Figure 2. Summary of Task Group 1 major tasks and subtasks.

TASK 1.1: Modernize and document DWRi's information model(s) for existing data and planned future data

GAPS ADDRESSED:

- **Gap 1:** Accumulated technical debt has made it difficult for DWRi to maintain or modify existing databases and software systems to meet operational needs.
- **Gap 2:** DWRi's existing software and database system architecture and design is not well documented.

- **Gap 3:** DWRi's existing software and database system architecture limits modularity, flexibility, and scalability.
- **Gap 4:** Data collection and data analysis/use are not sufficiently separated within DWRi's systems and workflows.
- **Gap 14:** DWRi does not maintain an auditable trail of documentation for data modifications/changes.
- **Gap 24:** DWRi's information model and data model implementations for the DVRT database were designed around older data collection workflows that no longer represent how DWRi operates and do not meet its current or future needs.
- **Gap 25:** DWRi's information and data models are not well documented.
- **Gap 26:** DWRi stores data of multiple types within the same database.
- **Gap 35:** Separation of accounting data and measurement data is insufficient.
- **Gap 38:** Limited capability for "calculated" data products.

Because information and data models provide a structured approach for understanding, communicating, and managing data requirements of a system, ensuring that information and data models are formalized and documented is foundational to the design and operation of software systems. As described in the Gap Analysis, we distinguish between a higher-level "information model" that describes the structure, behavior, and semantics of data versus a "data model," which is a more detailed and concrete representation of the information model focused on a particular functional use case (e.g., a data model for encoding data for storage within a relational database or a data model for encoding data in a file for transfer over the Internet). In practice, there are multiple physical implementations of the "information model," each of which can be considered a "data model."

The importance of understanding the data and metadata domain make this major task a high priority for implementation, thus it is prioritized in Task Group 1. While the Gap Analysis focused specifically on measurement data, Water Use Program data, and distribution accounting data, we recommend that DWRi consider the subtasks under this major task holistically across all of the datasets it manages. The following subtasks are recommended:

1. **Identify and describe the specific datasets that DWRi stores and manages**: The initial priority focus from the Gap Analysis was DWRi's measurement data (data retrieved from measurement stations in the field and from stations operated by partners), but we recommend that DWRi develop and maintain a comprehensive catalog of datasets/data products they maintain, including Water Use Program data, data produced by distribution accounting activities, water right information, etc. (Recommendation 40).
2. **Consider existing information/data models for use by DWRi:** Existing data models can inform DWRi's information model design for important datasets, including the Observations Data Model (ODM) and HydroServer's operational data model, which is an implementation of the Open Geospatial Consortium's SensorThings data model blended with ODM (Recommendation 5, 41).

3. **Design and document effective and flexible information model(s) for DWRi's measurement data, Water Use Program Data, and distribution accounting data:** The information models will describe the scope of the data they represent and will define the data, its relationships, and any rules governing it (e.g., required and optional elements and other constraints) (Recommendation 7, 42, 46, 58). Documentation will formalize information model design using UML, entity-relationship (ER) diagrams, or other methods for documenting information model design. We recommend that this documentation be captured in public repositories like GitHub.
4. **Design an effective versioning system for DWRi's data:** DWRi's existing data systems are inadequate for representing the different versions of data used in their operations. A robust versioning system that is accounted for in DWRi's information model design is needed, particularly for measurement data and distribution accounting data, to ensure that DWRi's information model(s) effectively distinguish between different data, different versions of data, and observed versus calculated data (Recommendation 27, 44, 58, 62).
5. **Develop and document physical data model implementations of the information model(s):** DWRi's operational databases and software systems will include multiple physical implementations of their information models, each of which needs to conform to the information model design (Recommendation 43). Each of these physical implementations is considered a data model. Examples of required data models include implementations for:
    a. Storage of data in a relational database management system.
    b. Transfer of data over the Internet in a machine-readable format (JSON schema).
    c. Transfer of data over the Internet as a comma separated values file (CSV).
6. **Deploy new database/data model implementation(s):** Each of the data models used by DWRi will need to be deployed within their operational software systems. Generally, this will be done as part of other tasks.
7. **Develop functionality for migrating existing data to the new database(s):** As DWRi implements new or modernized data models, existing data will need to be migrated from DWRi's existing databases into the updated systems. This process should be scripted for reproducibility and to ensure that data are transferred with fidelity. When migration is complete, DWRi should conduct a completeness and quality review to ensure that historical data were transferred with fidelity.
8. **Periodically review DWRi's data systems:** Given the volume and number of datasets already managed by DWRi along with new data requirements emerging with Utah's Agricultural Optimization Program, DWRi should regularly review (at least semi-annually) their data systems to ensure that all data encodings used by DWRi conform to the information model design(s) developed by this task and that the information models used by DWRi continue to meet emerging needs (Recommendation 6).

TASK 1.2: Establish software development processes and procedures

GAPS ADDRESSED:

- **Gap 1:** Accumulated technical debt has made it difficult for DWRi to maintain or modify existing databases and software systems to meet operational needs.
- **Gap 5:** Source code for DWRi's software applications is not stored or managed within shared source code management repositories.
- **Gap 6:** DWRi is not fully taking advantage of modern DevOps tools for software development, integration, and deployment.

The Gap Analysis identified opportunities for DWRi to adopt modernized software development tools and procedures to better facilitate co-development (i.e., collaborative software development by multiple developers within and outside of DWRi), to ensure that developed code is safely managed in a robust version control system, and to ensure that software development best practices are followed by existing and potential new employees, as well as by contributors outside of DWRi. This task is prioritized within Task Group 1 to ensure that established procedures are used in subsequent software development tasks. The following subtasks are recommended:

1. **Set up a GitHub Organization for DWRi:** GitHub is a widely used, robust source code management system with version control and support for multiple simultaneous developers. A GitHub Organization will enable DWRi to establish an online presence for designs, code, and documentation with supporting tools to help them with development and maintenance (Recommendation 9).
2. **Establish and operate source code repositories for important source code and software products:** Within a GitHub Organization, DWRi can establish individual code repositories for each software product (Recommendation 9). Repositories include version control for code, an issue tracker for tracking bugs and feature requests, formal tools for issuing releases and versioning code and software (Recommendation 13), and tools for developing documentation.
3. **Establish effective, shared, collaborative source code management practices among software developers:** Multiple DWRi employees contribute to the development of software used by DWRi. Additionally, DWRi will collaborate with others outside of their organization who will need to be able to contribute to code and documentation within DWRi's source code repositories. While the technical tools available through GitHub facilitate this, collaborative teams generally need to agree upon collaborative workflows and processes for effectively using these tools. DWRi's development best practices should be documented in a shared place available to all developers (Recommendations 10, 11, 12, 13).
4. **Establish effective code review procedures for collaborative software development work:** Code review is an important tool that can increase the quality of developed code through promoting consistency and compliance with best practices,

early detection of errors and bugs, and promotion of shared understanding among developers. DWRi should consider greater integration of code reviews in their software development processes (Recommendation 13).

TASK 1.3: Standardize measurement procedures

GAPS ADDRESSED:

- **Gap 1:** Accumulated technical debt has made it difficult for DWRi to maintain or modify existing databases and software systems to meet operational needs.
- **Gap 7:** Inconsistencies exist between DWRi and partner organizations in methods used for collecting and reporting data.
- **Gap 8:** DWRi lacks specifications for measurement hardware and measurement station inspection and maintenance visits.
- **Gap 9:** DWRi relies on "pull-based" data collection at many measurement stations that do not have local data storage.
- **Gap 10:** Many measurement stations are not automated, some data are not reported in a timely fashion, and some data are not available in a machine-readable format.
- **Gap 11:** The process for bringing new measurement stations online is not well documented.

The Gap Analysis described multiple gaps and recommendations related to standardizing, where possible, procedures used for making measurements of flow and other important observed variables to promote better consistency in data collected by DWRi and its partners who provide data. Data are most comparable and interoperable when they are collected using the same or similar methods, and steps made to improve the quality of data collected by DWRi and partners will carry through to enhancing the quality of and reducing uncertainty within analyses that use the data. We have also prioritized this task within Task Group 1 because, while it can be done independently of other major tasks, the sooner DWRi acts to promote greater quality and consistency in observed data, the sooner DWRi will realize the benefits of higher quality data. The following subtasks are recommended:

1. **Regularly assess existing flow measurement devices:** Establish a program for reviewing and assessing existing flow measurement devices that are already installed in the field for conditions that may impact accuracy, including age, installation, maintenance, etc. Conditions may change over time, so this program should be ongoing.
2. **Develop specifications and standardized procedures for data collection and reporting:** Promote consistency in collected data through specifying which variables are recorded for operational purposes, observation recording intervals, and observation time support (Recommendation 15).
3. **Develop specifications and standards for measurement station hardware and programming:** Develop specifications for acceptance of flow measurement devices to

be installed at measurement stations, including appropriateness for station conditions, required accuracy, etc. Provide guidelines on how stations should be programmed to ensure that consistent data are reported (Recommendation 17).

4. **Establish a program to systematically move toward automated and telemetered measurement stations:** DWRi should continue existing programs for automating and telemetering measurement stations (Recommendation 21) with a goal of reducing the number of stations where data are not reported in a timely way or where data are not reported in a machine-readable format, including digitizing historical measurement data available in PDF and/or other formats for priority distribution systems (Recommendation 22). We also recommend that DWRi replace legacy, on-demand, "pull-based" data collection systems with modernized dataloggers and telemetry (Recommendation 20).
5. **Develop requirements and protocols for measurement station visits:** Develop program requirements for how often measurement stations should be visited, who should conduct station visits, and a protocol for assessing site conditions that may impact the quality and accuracy of measurements made by that station (Recommendation 18).
6. **Develop documentation of standardized procedures: Develop** documentation of items 2–4 as one or more "Techniques and Methods" reference documents that can be used as a reference for anyone who is deploying new measurement stations to report data to DWRi (Recommendation 19).

TASK GROUP 2: MODERNIZE CORE SOFTWARE SYSTEMS

The Gap Analysis identified several areas where the core operational software systems currently in use by DWRi for data collection and management are in need of enhancement and/or modernization. After many years of successful operation, the level of technical debt inherent in DWRi's existing system and the associated difficulties they have in maintaining, operating, and upgrading their existing system led USU to recommend significant updates to or replacement of several operational systems DWRi currently relies upon for day-to-day work. A significant challenge in modernizing these core systems will be updating the "downstream" software applications that are dependent upon the core systems. Thus, we anticipate that DWRi will need to operate their existing systems (e.g., the DVRT database and various data loaders) in parallel with newly developed/implemented systems until they establish confidence that the new systems have fully captured the data and functionality they need, and that downstream use of the data hosted by those systems has been successfully reconfigured. Each of the major tasks prioritized in this task group, along with recommended subtasks, are summarized in Figure 3 and described in the following sections.

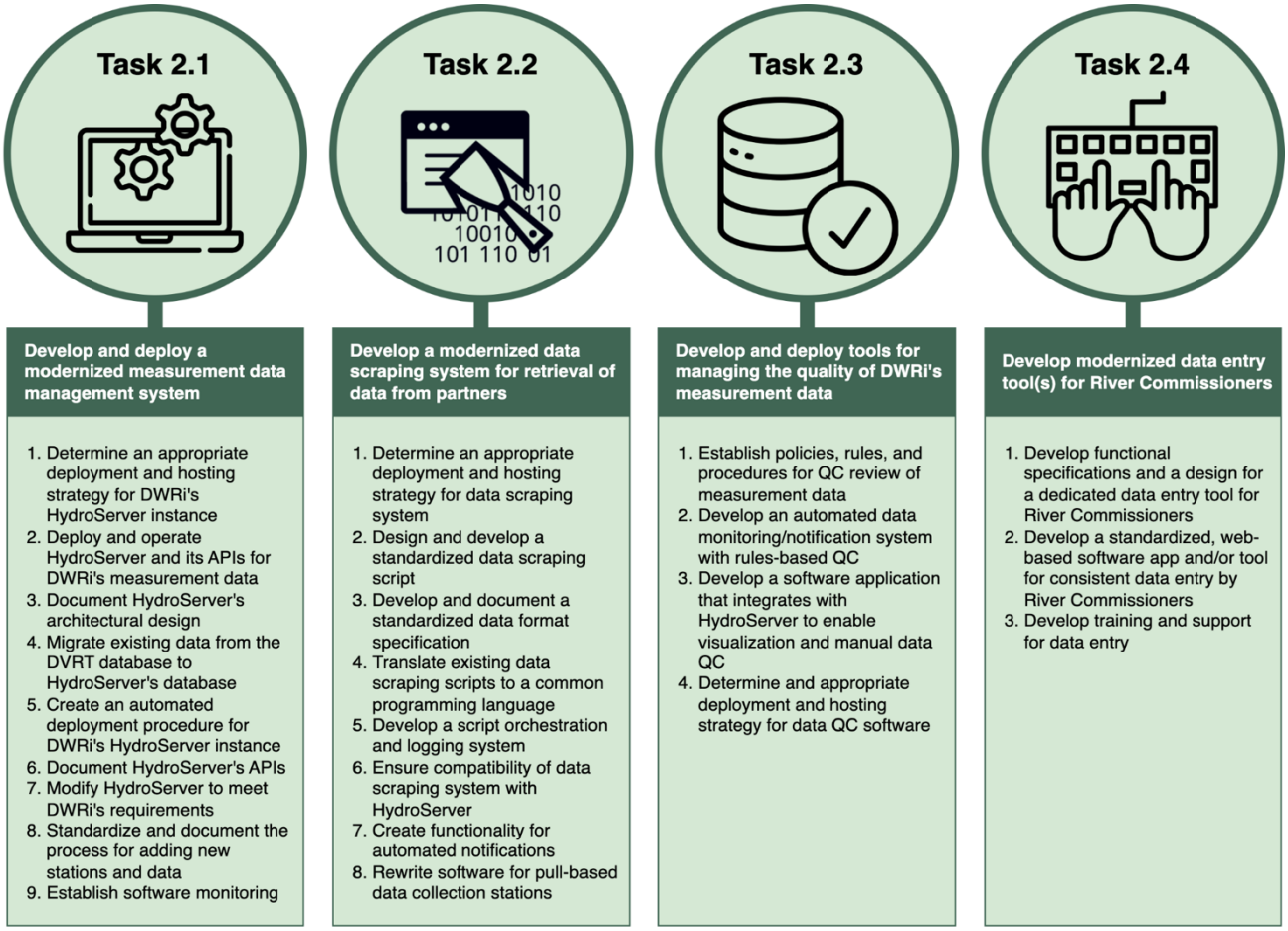


Figure 3. Summary of Task Group 2 major tasks and subtasks.

TASK 2.1: Develop and deploy a modernized measurement data management system

GAPS ADDRESSED:

- **Gap 1:** Accumulated technical debt has made it difficult for DWRi to maintain or modify existing databases and software systems to meet operational needs.
- **Gap 3:** DWRi's existing software and database system architecture limits modularity, flexibility, and scalability.
- **Gap 4:** Data collection and data analysis/use are not sufficiently separated within DWRi's systems and workflows.
- **Gap 12:** Existing measurement stations lack descriptive metadata.

- **Gap 14:** DWRi does not maintain an auditable trail of documentation for data modifications/changes.
- **Gap 24:** DWRi's information model and data model implementations for the DVRT database were designed around older data collection workflows that no longer represent how DWRi operates and do not meet its current or future needs.

This task is aimed at developing a modernized system for data collection and management that reduces technical debt (Recommendation 1), is fully independent from DWRi's applications of the data (i.e., distribution accounting) to ensure that field observations and calculated or derived data products are kept separate and adequately described with metadata (Recommendation 7). After considering several options in the Gap Analysis (Recommendation 4), DWRi has indicated a preference for deploying an instance of the HydroServer (https://www.hydroserver.org) software stack, which implements a modern information model (Recommendation 5), has a modular services-oriented architecture (Recommendation 3), and supports multiple measurement types and versioning for time series of measurement data (Recommendation 27). HydroServer is a software stack based on the Open Geospatial Consortium's (OGC) SensorThings API and data model that "out-of-the-box" provides much of the functionality DWRi needs for their measurement data management system. A major motivator for using HydroServer is that it can be modified to meet DWRi's specific needs for collecting, storing, managing, and sharing measurement data. This task is grouped with Task Group 2 because it relies on information modeling results and established software development procedures set in Task Group 1. The following subtasks are recommended:

1. **Determine an appropriate deployment and hosting strategy (platforms and technologies) for DWRi's HydroServer instance:** HydroServer is a software stack that includes a time series database, web service APIs, and web applications that enable collection, storage, and management of time series of observational data. HydroServer's preferred deployment environment is cloud-based using Amazon Web Services (AWS), but HydroServer's software components can be deployed using infrastructure on-premises or in the cloud. DWRi will need a deployment and hosting strategy for their HydroServer instance that is consistent with any state IT rules and procedures (Recommendation 10, 11, 12).
2. **Deploy and operate an instance of HydroServer and its application programming Interfaces (APIs) for DWRi's measurement data:** Deployment of the HydroServer software stack using AWS is straightforward using instructions and tools provided by the HydroServer development team. If DWRi wants/needs to host any of the HydroServer components on-premises, additional deployment strategies and documentation will need to be developed. HydroServer includes APIs for data ingest into its operational database, for data query/retrieval, and for modifying metadata within the system. These APIs are deployed along with the HydroServer software stack (Recommendation 31, 50).
3. **Document the architectural design of the system and its major components:** DWRi will need documentation of HydroServer's structure, components, and deployment

configurations to ensure successful operation and maintenance and to ensure integration with DWRi's other operational software components (e.g., distribution accounting) (Recommendation 2).

4. **Migrate existing data from the DVRT database to HydroServer's operational database:** All of DWRi's existing measurement data will need to be copied to their operational HydroServer instance to ensure continuity of operations. This process should be scripted for efficiency and to minimize opportunities for errors and may require work to annotate existing measurement stations with additional metadata where available (Recommendation 24).
5. **Create an automated deployment procedure for maintaining DWRi's HydroServer instance(s):** DWRi will have a primary production HydroServer instance and may choose to set up additional instances for prototyping development and testing bug fixes, enhancements, and new functionality (Recommendation 10). DWRi may want to set up an automated deployment procedure using tools available in GitHub and designed to automate the procedure for updating DWRi's HydroServer instance(s) to the latest testing or release versions (Recommendation 11). Automated testing of software and deployments is also recommended (Recommendation 12).
6. **Develop documentation and code examples for interacting with HydroServer APIs:** Developing software applications that insert data into DWRi's HydroServer instance or that query data out of HydroServer's database will require developing code that uses HydroServer's APIs. Basic documentation for HydroServer's APIs already exists, but DWRi may need to develop additional, agency-specific documentation for how HydroServer's APIs are used operationally to support DWRi's software applications (e.g., software that loads data from measurement stations or software that retrieves and uses measurement data in distribution accounting). DWRi may also choose to provide documentation for users outside of DWRi on how to use available APIs (Recommendation 2).
7. **Modify DWRi's HydroServer instance to meet DWRi's requirements:** HydroServer is a generic software stack for managing data from environmental sensors. DWRi will likely want to modify their HydroServer instance to control the branding and look and feel of the user interface and to implement specific functional requirements identified through examination of DWRi's use cases.
8. **Standardize and document the process for adding new measurement stations, observed variables, and data values to DWRi's HydroServer instance:** DWRi will want to develop protocols around who can add new measurement stations, observed variables, and new data values to DWRi's HydroServer instance (Recommendation 23). This is both a policy and access control issue.
9. **Establish appropriate monitoring of software application health to enhance reliability:** DWRi will want to consider deploying automated tools for monitoring the health and reliability of their HydroServer instance (and potentially other web applications). These applications can provide useful notifications if/when applications go down to ensure they are fixed promptly. They can also provide useful metrics for reporting (e.g., uptime) (Recommendation 12).

TASK 2.2: Develop a modernized data scraping system for retrieval of data from partner providers

GAPS ADDRESSED:

- **Gap 1:** Accumulated technical debt has made it difficult for DWRi to maintain or modify existing databases and software systems to meet operational needs.
- **Gap 3:** DWRi's existing software and database system architecture limits modularity, flexibility, and scalability.
- **Gap 9:** DWRi relies on "pull-based" data collection at many measurement stations that do not have local data storage.
- **Gap 16:** DWRi relies on a large number of non-standardized data scraping scripts to retrieve data from partners.
- **Gap 17:** DWRi could enhance orchestration and monitoring of data scraping scripts to improve data integrity and performance.
- **Gap 18:** DWRi's existing methods for data scraping and retrieval of data from different sources may not scale to meet upcoming data management needs.

The Gap Analysis identified DWRi's data scraping scripts as an area for coordinated updates that can help DWRi move toward more standardized methods and protocols for retrieving data and systems for operating data scrapers to improve reliability. DWRi currently relies on many different data scraping scripts to retrieve measurement data from many different sources. While DWRi will likely have to rely on data scraping for the foreseeable future until the capabilities of data providers can be increased, addressing the inherent fragility of the current set of data scraping scripts can be accomplished through better orchestration and standardization (Recommendation 30). Additionally, as DWRi deploys new data management tools (e.g., the ability for data providers to push data into DWRi's HydroServer instance rather than DWRi having to pull the data) this will provide opportunities for promoting a greater level of sophistication from data providers and moving away from data scraping in favor of partners pushing data to DWRi. This task must follow Task 2.1 as the target for the data loading tasks accomplished by modernized data scraping scripts will be DWRi's HydroServer instance. The following subtasks are recommended:

1. **Determine an appropriate deployment and hosting strategy (platforms and technologies) for DWRi's data scraping system:** DWRi's data scraping scripts have computational needs and must be scheduled as jobs that run regularly. A strategy is needed for hosting the orchestration system that runs and monitors data scraping scripts either in the cloud or via an on-premises physical or virtual machine acting as an application server (Recommendation 10, 11, 12).
2. **Design and develop a standardized data scraping script and ability to run additional scripts where needed:** While partners currently provide data in many different file formats and syntax, if data were provided in a consistent format, a single

data scraping script could be designed to load data from any data sources using that format. DWRi should implement a standardized data scraping script that can load data from any provider that uses DWRi's standardized data format specification (Recommendation 30). We recognize that some data providers will not or may not be able to adopt DWRi's standardized data format specification immediately, so it will be necessary for DWRi's script orchestration system to run additional scripts beyond the standardized data scraping script.

3. **Develop and document a standardized data format specification:** A standardized data scraping script (subtask 2) requires a standardized data format specification for data to be scraped (Recommendation 30). This format should be easily retrieved, read, parsed, and validated for loading data (e.g., likely a simple CSV file format).
4. **Translate existing data scraping scripts to a common programming language:** DWRi has developed data scraping scripts over time and using multiple programming languages. Maintaining scripts in different languages is an unnecessary complexity that could be removed by standardizing all scripts to use a single language (likely Python) (Recommendation 30).
5. **Develop an orchestration and logging system that runs data scraping scripts:** A system is needed that runs data scraping scripts on a user-specified schedule and logs useful results and diagnostic information to assist with monitoring operations and debugging. Such a system would help in improving the reliability of DWRi's data scraping operations (Recommendation 30).
6. **Ensure compatibility of modernized data scraping system with HydroServer:** The target for DWRi's data scraping scripts to load new data will be DWRi's HydroServer instance. Thus, existing and new data scraping scripts will need to be (re)configured to use HydroServer's APIs to load data (Recommendation 31). If scripts are written in Python, they can use HydroServer's Python client package to facilitate simpler data loading.
7. **Create functionality for automated notifications:** Data scraping operations are inherently fragile given that DWRi does not control what happens on the data provider end when pulling data. Data sources may change the location, names, or formats of files containing data, which will break scraping operations. DWRi's script orchestration system (subtask 5) should be enhanced to send alerts to data managers when data scraping is unsuccessful or when errors are encountered to ensure that errors can be addressed quickly (Recommendation 30).
8. **Rewrite software that retrieves data from legacy, on-demand, "pull-based" data collection stations:** DWRi relies on legacy Fortran code to retrieve data from their older "pull-based" data collection stations. Given that these stations will be in operation for the foreseeable future, DWRi should rewrite this functionality in a modern language (e.g., Python for consistency) to load data from these stations automatically into DWRi's HydroServer instance (Recommendation 20).

TASK 2.3: Develop and deploy tools for managing the quality of DWRi's measurement data

GAPS ADDRESSED:

- **Gap 1:** Accumulated technical debt has made it difficult for DWRi to maintain or modify existing databases and software systems to meet operational needs.
- **Gap 13:** DWRi lacks specific policies, rules, and procedures for reviewing observed data and data quality control.
- **Gap 14:** DWRi does not maintain an auditable trail of documentation for data modifications/changes.
- **Gap 15:** DWRi lacks sufficient tools and personnel for reviewing data quality.
- **Gap 35:** Separation of accounting data and measurement data is insufficient.

The Gap Analysis identified several gaps related to data quality control (QC) for measurement data retrieved from stations operated by DWRi and partner organizations. Given the number of measurement stations and the volume of data DWRi collects and manages, a suite of manual and automated QC tools would help DWRi identify data quality issues and then correct them where necessary to better manage and ensure the quality of the data used in DWRi's operational work (e.g., distribution accounting). This task must follow Task 2.1 because tools developed for data QC should interact directly with data stored in DWRi's HydroServer instance. The following subtasks are recommended:

1. **Establish policies, rules, and procedures for QC review of measurement data and potentially other important datasets:** As part of DWRi's overall data governance strategy, specific policies, rules, and procedures should be developed to determine who is responsible for data QC, how they should do it, and how results should be managed—e.g., a strategy for data versioning and auditing of modifications (Recommendation 25). Training of employees will also be required (Recommendation 26).
2. **Develop an automated data monitoring/notification system that integrates with HydroServer and implements automated, rules-based quality assurance/QC for DWRi's measurement data:** Rules-based QC may include automated range checks, persistence checks, missing data checks, battery voltage checks, etc. necessary to avoid unnecessary data loss at DWRi's measurement stations. The large number of stations with active data collection and the volume of data produced will preclude a human from regularly visualizing and analyzing incoming data. Automated rules and notifications can help data managers focus on specific problem areas (Recommendation 28).
3. **Develop a software application that integrates with HydroServer to enable data visualization and manual data QC for DWRi's data managers based on policies provided by DWRi:** This software application may be developed as a web application or a desktop client application that enables data managers to visualize data, identify anomalies, and flag and/or correct them using simple tools (Recommendation 26).

4. **Determine an appropriate deployment and hosting strategy (platforms and technologies) for data quality control software tools:** An automated, rules-based QC notification system would have computational needs that could be met using cloud or on-premises infrastructure. Depending on how a manual QC application is built (i.e., as a desktop client or web application), it may also need an appropriate strategy for deployment and hosting.

TASK 2.4: Develop modernized data entry tool(s) for River Commissioners

GAPS ADDRESSED:

- **Gap 1:** Accumulated technical debt has made it difficult for DWRi to maintain or modify existing databases and software systems to meet operational needs.
- **Gap 19:** DWRi relies heavily on manual data entry.
- **Gap 35:** Separation of accounting data and measurement data is insufficient.
- **Gap 40:** Insufficient technical skills among River Commissioners.

DWRi currently relies heavily on manual data entry by River Commissioners. While the Gap Analysis recommended that DWRi move away from manual data entry as much as possible, some manual data entry may continue to be unavoidable given that many Distribution Systems have measurement stations that are not automated or do not have telemetry. Existing tools for data entry are functional, but do not meet all existing needs. Better tools are needed to enhance data collection efficiency, timeliness of reporting, and the accuracy of manual data entry by River Commissioners (Recommendation 33). The following subtasks are recommended:

1. **Develop functional specifications and a design for a dedicated data entry tool for River Commissioners:** Engage River Commissioners directly to solicit their ideas for required functionality and develop a design and functional specifications for a dedicated data entry tool that will better meet their needs.
2. **Develop a standardized, web-based software app and/or tool for consistent data entry by DWRi's River Commissioners:** Based on the functional specifications developed in subtask 1, develop a data entry tool that is compatible with DWRi's HydroServer instance (i.e., loads data directly into HydroServer). This may be a mobile device application, a Progressive Web App (PWA), or adaptation of an existing application to support manual data entry.
3. **Develop training and support for data entry**: Develop training and support materials (e.g., documentation) to enable River Commissioners to transition to new data entry tool(s) to reduce the potential for errors in data entry, and to ensure effectiveness.

TASK GROUP 3: APPLY MODERNIZED SOFTWARE

DWRi's measurement data management system is the foundation on which they build other analyses that use the measurement data. Thus, downstream applications that use the data must be updated to interface with DWRi's modernized measurement data system - e.g., they will need to be modified to retrieve data from HydroServer's API rather than interfacing directly with a relational database and they may need to be modified to store results back to DWRi's HydroServer instance or another database maintained by DWRi. Task Group 3 includes those tasks that must be completed after the new measurement data management system has been implemented, after existing measurement data has been migrated, and after any operational data loaders/data scrapers have been reconfigured to load data to the new measurement data management system. Each of the major tasks prioritized in this task group along with recommended subtasks are summarized in Figure 4 and described in the following sections.

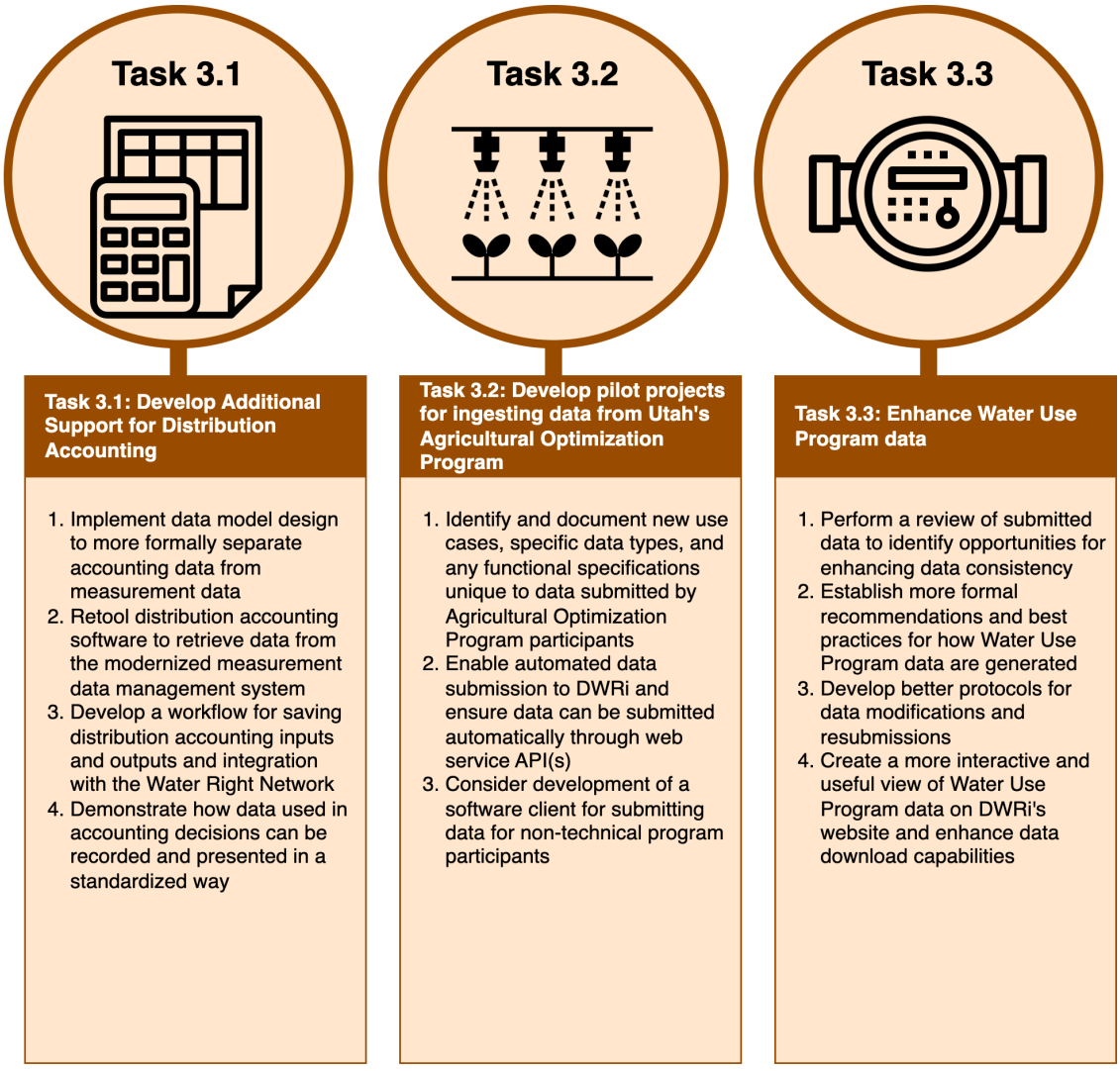


Figure 4. Summary of Task Group 3 major tasks and subtasks.

TASK 3.1: Develop additional support for distribution accounting

GAPS ADDRESSED:

- **Gap 1:** Accumulated technical debt has made it difficult for DWRi to maintain or modify existing databases and software systems to meet operational needs.
- **Gap 35:** Separation of accounting data and measurement data is insufficient.
- **Gap 36:** Inconsistent and complex presentation of water rights accounting information across distribution systems.
- **Gap 37:** Some processes in distribution accounting and allocation decision making are manual.
- **Gap 38:** Limited capability for "calculated" data products.

Distribution accounting is one of the primary uses of DWRi's measurement data. The Gap Analysis found inconsistent separation between measurement data and distribution accounting modeling/results such that the data that were used or the modifications that were made to the data prior to input to accounting models were not always transparent. Furthermore, in efforts to increase the transparency of decision making, DWRi wants to save a record of the data used as inputs and the results of distribution accounting. This will require careful consideration of how to version data for clear provenance and how to represent, store, and share distribution accounting data. The following subtasks are recommended:

1. **Implement data model design to more formally separate accounting data from measurement data in DWRi's data storage infrastructure:** As part of Task 1.1, DWRi should consider the information model needed to represent distribution accounting data—inputs and outputs to DWRi's distribution accounting—separately from measurement data (Recommendation 7, 58). DWRi's HydroServer instance and associated database may be adequate for storing and sharing these data as is; however, given the accounting context, the desire to store "calculated" data products (e.g., measurement stations representing the sum of two diversions), and potentially additional required metadata indicating different data versions, additional functionality specific to the distribution accounting context may be required. This subtask acknowledges that the information modeling work in Task 1.1 may require iterative design to ensure representation, storage, and exchange of distribution accounting inputs and outputs.
2. **Retool distribution accounting software to retrieve data from the modernized measurement data management system via API:** DWRi's existing tools for distribution accounting modeling retrieve data from DWRi's DVRT database as inputs and then allow operators to review and modify those inputs. These software tools will need to be modified to retrieve data instead from DWRi's HydroServer instance via API, and, where input data are modified in the distribution accounting process, potentially write data back to HydroServer as a separate version of the data used for accounting purposes.

3. **Develop a workflow for saving distribution accounting inputs and outputs and for supporting integration of distribution accounting inputs and outputs with the Water Right Network:** One of the major goals of the DWRi's Water Right Network is to integrate hydrographic features, points of diversion, water right information, and areas of use with time series of flow measurement data and distribution accounting data/results. A workflow is needed to first save the inputs and outputs of distribution accounting (Recommendation 61) and to integrate those data with the geospatial features of the Water Right Network to ensure that they can be efficiently integrated and presented to potential users.
4. **Demonstrate how data used in accounting decisions can be recorded and presented in a standardized way:** The Gap Analysis found inconsistent and complex presentation of water rights accounting information across distribution systems. This inconsistency and complexity make the details of accounting and its outputs unapproachable for many prospective users and could be addressed by designing specific user-interfaces for presenting the data to specific stakeholders (e.g., web-based "dashboards" or other data and distribution visualization tools aimed at DWRi's Distribution Engineers, River Commissioners, or the general public). DWRi should develop one or more pilot application(s) for distribution system(s) that have good infrastructure to support it (e.g., high-quality measurement data, storage accounts, complete representation of the geospatial data used in the Water Right Network, etc.) (Recommendation 59). These pilots could then be used to elicit feedback from stakeholders to help DWRi refine the design before developing similar interfaces for all distribution systems in the state. These dashboards may also be used to demonstrate how distribution accounting can be integrated with the Water Right Network (Recommendation 60).

TASK 3.2: Develop pilot demonstration projects for ingesting data from Utah's Agricultural Optimization Program projects

GAPS ADDRESSED:

- **Gap 1:** Accumulated technical debt has made it difficult for DWRi to maintain or modify existing databases and software systems to meet operational needs.
- **Gap 18:** DWRi's existing methods for data scraping and retrieval of data from different sources may not scale to meet upcoming data management needs.
- **Gap 41:** Coordination between DWRi and State of Utah DTS could be enhanced to better meet DWRi's technology needs.

Utah's Agricultural Optimization Program represents a significant new source of data that DWRi will need to manage. Projects are now coming online with the requirement to submit data to DWRi, but there is uncertainty about what data should be submitted and how it should be used to assess the effectiveness of funded projects with respect to water conservation. Thus, under this major task, we recommend that DWRi conduct a set of pilot projects to develop

requirements, procedures, and standards for program participants to submit data (Recommendation 32). This task is part of Task Group 3 because it will rely on the data management infrastructure put in place under Task Group 2 for enabling submission, storage, management, and analysis of submitted data. The following subtasks are recommended:

1. **Identify and document new use cases, specific data types, and any functional specifications unique to data to be submitted by Agricultural Optimization Program participants:** Data submitted by program participants may include time series data from meters and other flow measurement devices, which can likely be handled by DWRi's HydroServer instance. However, other types of data may need to be submitted. Scoping and documentation of data to be submitted is required to ensure that submitted data are compatible with DWRi's existing software systems or that information and data models are expanded to accommodate the data DWRi needs to manage for this program. This scoping can be done through developing pilot projects that examine use cases from early program participants.
2. **Enable automated data submission to DWRi and ensure data can be submitted automatically through DWRi's web service API(s):** Given the number of funded projects and the scale of potential data to be submitted by program participants, it would be ideal for data submission from program participants to be automated. Under this subtask, DWRi would partner with pilot project participants (i.e., early participants of the Agricultural Optimization Program who have access to significant technical expertise to test the functionality of HydroServer's APIs for data submission and would identify data and use cases for which automated data submission cannot be accomplished using existing APIs. These use cases could then be prioritized for further software development.
3. **Consider development of a software client for submitting data for non-technical program participants:** Some program participants may lack sufficient technical expertise or programmatic support to establish automated data submission. For these users, a client software application that guides them through the process of data submission, validates the data they provide, and manages submission to DWRi's systems may be useful. Scoping for such a client program would be based on early pilot demonstration projects.

TASK 3.3: Enhance Water Use Program data

GAPS ADDRESSED:

- **Gap 1:** Accumulated technical debt has made it difficult for DWRi to maintain or modify existing databases and software systems to meet operational needs.
- **Gap 20:** Inconsistencies in self-reported data lead to challenges for DWRi in combining data into a consistent dataset.
- **Gap 21:** A significant portion of self-reported data are based on estimates and not measurements.

- **Gap 23:** Navigating, downloading, and interpreting Water Use Program data on DWRi's website is difficult.

The Water Use Program Database is an important dataset assembled by DWRi and used by multiple partner organizations. The Gap Analysis identified several opportunities for improving the reliability and consistency of data submitted to this program database through improving the data submission process and through minimizing the amount of data processing and management DWRi must do before sharing a consistent and reliable dataset to which stakeholders have access. This task follows the information modeling work in Task Group 1, which will revisit how the Water Use Program data are stored and represented as a first step in enhancing the Water Use Program data. The following subtasks are recommended:

1. **Perform a review of submitted data to identify opportunities for enhancing data consistency:** DWRi could perform a review of Water Use Program Data to identify areas where water users are inconsistently reporting information (Recommendation 34). These areas could then be targeted for additional documentation guidance, and potentially even automated technical data validation upon submission to promote greater consistency across data providers. This could also include a review of data integrity for water transfers between systems to avoid double counting withdrawals. DWRi may also consider adding visualization of time series data per system and water source (see subtask 4 below) to show any potential anomalies.
2. **Establish more formal recommendations and best practices for how Water Use Program data are generated:** Development of documented recommendations and best practices around opportunities/practices for enhancing data consistency identified in subtask 1 and other specific guidance for data collectors and submitters, including specific guidance on measuring devices and metadata would enhance the quality and consistency of reported data (Recommendation 36).
3. **Develop better protocols for data modifications and resubmissions:** The Gap Analysis identified a need for better supporting data submitters in modifying or correcting the data they have submitted after the initial submission (Recommendation 35). This would require modifications to the data submission process to allow edits to prior submissions (potentially even for prior years' reports) and potentially additional metadata to help track resubmissions.
4. **Create a more interactive and useful view of Water Use Program data on DWRi's website and enhance data download capabilities:** The current view of a water supplier's information is a single, very long page with no interactive elements that requires significant scrolling. A more interactive view that allows users to "drill-down" into specific sections of the data might help users more easily find and access specific information in the reported data (Recommendation 39). Additionally, greater integration of the website view of the data with download capabilities and enhanced metadata for interpreting data downloads would facilitate easier access (Recommendation 37, 38).

TASK GROUP 4: ADVANCE THE WATER RIGHT NETWORK

The Water Right Network is a geodatabase that incorporates geographic features representing Utah's hydrography, points of diversion, diversion measurement locations, places of use for water rights, observed data from diversions, water right information, and distribution accounting results. As a comprehensive geodatabase, it requires an organized information model and physical implementations of that information model to enable storing and exchanging the data. While elements of the Water Right Network can be developed independently (e.g., the geometry of the hydrographic features), full implementation of the Water Right Network is dependent upon results of the first three Task Groups, and so we have specified Water Right Network tasks within Task Group 4. Each of the major tasks prioritized in this task group, along with recommended subtasks, are summarized in Figure 5 and described in the following sections.

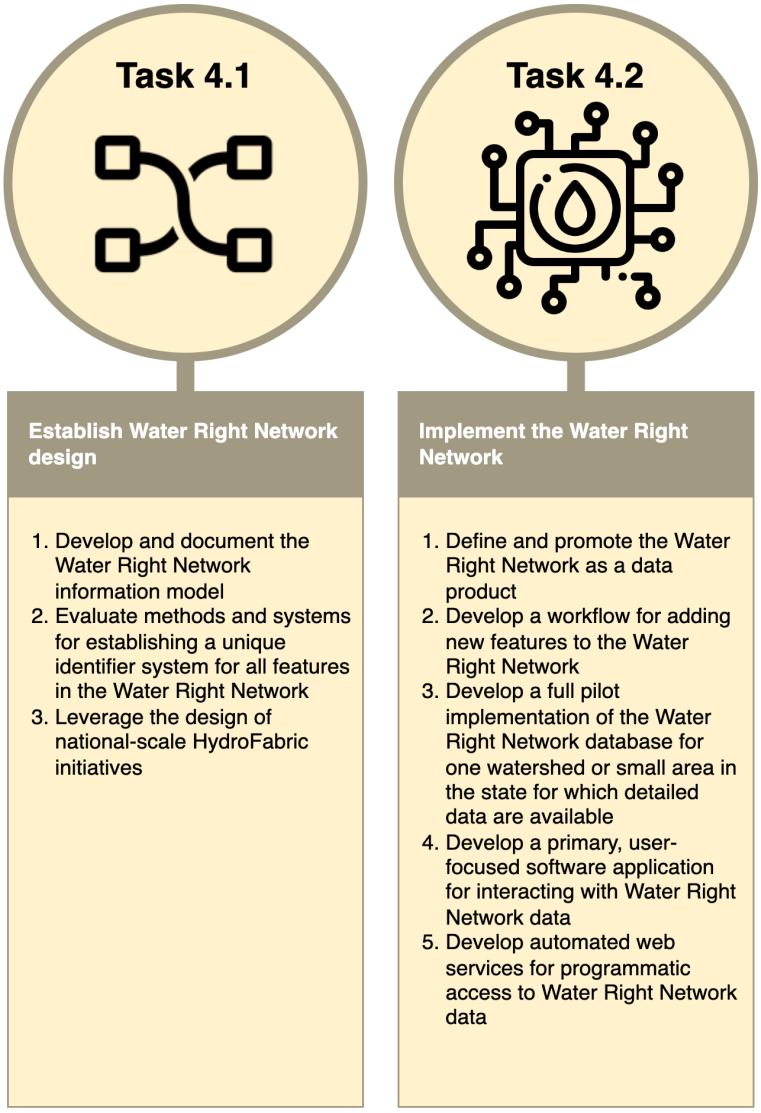


Figure 5. Summary of Task Group 4 major tasks and subtasks.

TASK 4.1: Establish Water Right Network design

GAPS ADDRESSED:

- **Gap 24:** DWRi's information model and data model implementations for the DVRT database were designed around older data collection workflows that no longer represent how DWRi operates and do not meet its current or future needs.
- **Gap 30:** It is not yet clear to potential users what the Water Right Network is.
- **Gap 31:** Design documentation for the Water Right Network data model design is not accessible.
- **Gap 32:** DWRi's identification system for geospatial entities may be inadequate.
- **Gap 35:** Separation of accounting data and measurement data is insufficient.
- **Gap 36:** Inconsistent and complex presentation of water rights accounting information across distribution systems.

Given that the Water Right Network is fundamental to DWRi's future plans for distribution accounting and for providing access to all of the datasets that are part of the Water Right Network, documentation of the information model design and its implementation is important for developers and users. Additionally, the Water Right Network is an innovative idea that can serve as an example for how other states/jurisdictions could organize data to support its understanding and use. Thus, documentation of the Water Right Network design also serves to communicate this innovative work outside of DWRi. The following subtasks are recommended:

1. **Develop and document the Water Right Network information model:** While Task 1.1 focuses on information model design, we include this subtask here because the Water Right Network includes multiple types of data and thus must include the information models of the incorporated data. The information model for the Water Right Network and its physical implementation as a geodatabase must specify comprehensive metadata, determine how features are related, and show how each of the datasets is linked. Documentation of the design and implementation is critical for effective implementation, maintenance, and use (Recommendation 52).
2. **Evaluate methods and systems for establishing a unique identifier system for all features in the Water Right Network:** The complexity and number of features involved in the Water Right Network requires a robust approach for establishing globally unique feature identifiers that can be used to uniquely identify features and to link features and their attributes via topological relationships. Robust identifiers ensure data uniqueness across systems, which helps maintain data integrity and avoid conflicts when data are moved or exchanged (Recommendation 55). Use of identifiers may need to be a broader, more holistic consideration across DWRi's data systems given that several of DWRi's datasets will participate in the Water Right Network.
3. **Leverage the design of national-scale HydroFabric initiatives**: Multiple, national-scale efforts to develop "HydroFabrics" for modeling purposes are underway by the

United States Geological Survey (USGS) and by the National Oceanic and Atmospheric Administration (NOAA). These HydroFabrics are hydrologically connected hydrographic networks on which national scale models are run with associated software tools and methods that may be useful for DWRi's Water Right Network HydroFabric design and development (Recommendation 54).

TASK 4.2: Implement the Water Right Network

GAPS ADDRESSED:

- **Gap 33:** Limited public accessibility and usability of current data presentation.
- **Gap 34:** Lack of automated data access points.
- **Gap 37:** Some processes in distribution accounting and allocation decision making are manual.

DWRi seeks to use the Water Right Network as a means for providing access to geographically referenced data, as well as to serve as the eventual basis on which water distribution accounting could be based. Implementation of the Water Right Network will have a strong effect on how it is used by stakeholders within and outside of DWRi. The following subtasks are recommended:

1. **Define and promote the Water Right Network as a data product:** We recommend that DWRi develop visual examples, illustrations, design materials, documentation, etc. to help communicate the purpose, design, and use of the Water Right Network to all potential stakeholders (Recommendation 51).
2. **Develop a workflow for adding new features to the Water Right Network:** New diversion locations, measurement stations, etc. created by DWRi to address gaps in measurements or modifications to points of diversion or use will need to be added to the Water Right Network. This will require a specific protocol and workflow for ensuring that necessary features are modified or added, that they are linked correctly to the hydrographic network, and that they are linked to associated water right information, observational data, and/or distribution accounting results (Recommendation 45).
3. **Develop a full pilot implementation of the Water Right Network database for one watershed or small area in the state for which detailed data are available:** A tremendous amount of work is required to implement the Water Right Network statewide. While this work is being completed, we recommend that DWRi develop a full pilot implementation of the Water Right Network for one distribution system or one priority area of the state that can be used for demonstrations, development of data management workflows, prototyping of distribution accounting based on the Water Right Network, and development of documentation materials (Recommendation 53).
4. **Develop a primary, user-focused software application for interacting with Water Right Network data:** For users to be able to fully understand the Water Right Network, DWRi needs to be able to point users at a primary user interface that can be used to

browse and interact with the Water Right Network data. This software application should be web-based, include a map interface for viewing the geospatial data, and may include functionality for users to access, plot, or download parts of the Water Right Network data for their location of interest (Recommendation 56). Functional specifications and a design for this application should be based on user needs.

5. **Develop automated web services for programmatic access to Water Right Network data:** DWRi and its stakeholders may require automated access to Water Right Network data for several reasons, including automated distribution accounting based on interaction with the Water Right Network, automation of maintenance tasks such as adding new features, and querying subsets of the data for specific areas of the state (Recommendation 50, 57). Automated web services would facilitate these types of applications that might be constrained if the only access mechanism is a graphics user interface (subtask 4).

TASK GROUP 5: ENGAGE STAKEHOLDERS AND IMPLEMENT FEEDBACK

In Figure 1, the task area related to engaging stakeholders and implementing feedback is connected to each of the other task areas. We have not assigned this task area a sequence number, rather the connections to each other task area imply that DWRi should identify stakeholders (both within and outside of DWRi) to provide feedback during each major step of the Technology Modernization Roadmap to ensure that modifications, modernization, and addition of new technologies is done in a way that meets DWRi's and other stakeholders' needs. While we make some specific suggestions here based on gaps/recommendations from the Gap Analysis document, we recommend that stakeholder engagement and user-centered design be fundamental to DWRi's software development and implementation process, regardless of whether code is developed by DWRi or an external partner. Each of the major tasks prioritized in this task group, along with recommended subtasks, are summarized in Figure 6 and described in the following sections.

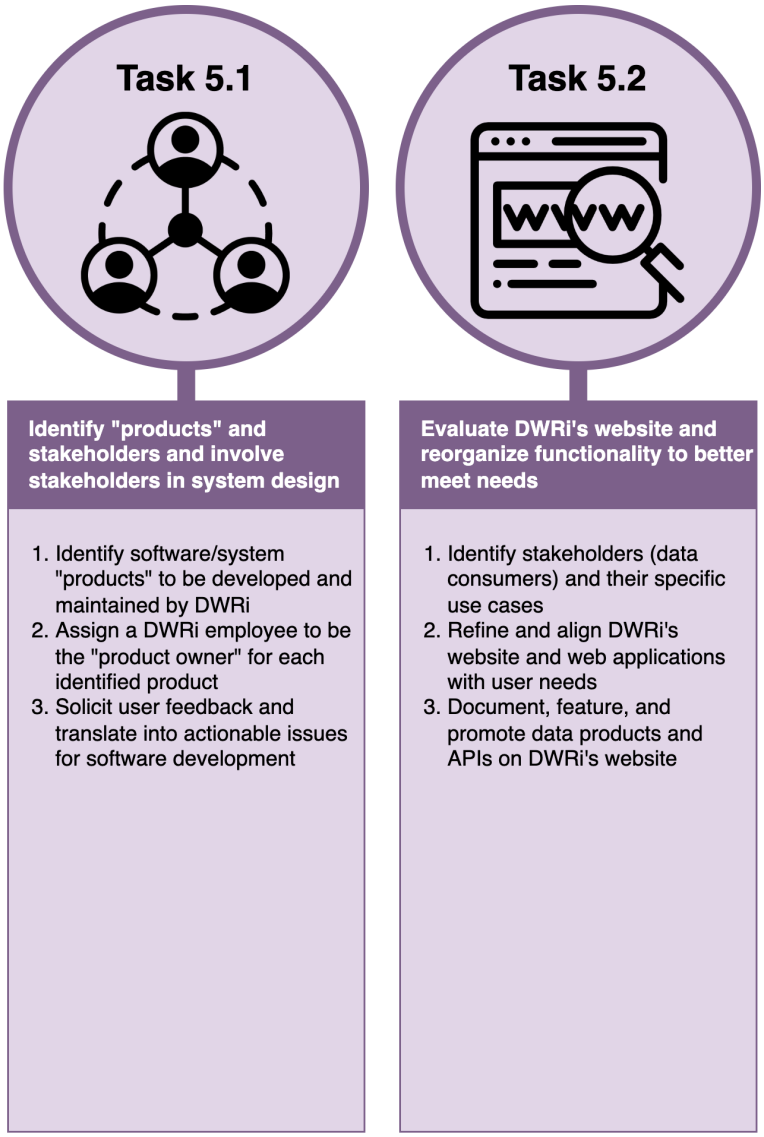


Figure 6. Summary of Task Group 5 major tasks and subtasks.

TASK 5.1: Identify "products" and stakeholders and involve stakeholders in design and implementation of DWRi's modernized data systems

GAPS ADDRESSED:

- **Gap 39:** Lack of dedicated data management roles and personnel.

To ensure that DWRi's software systems meet current and future data management needs, it will be important for DWRi to identify specific users/stakeholders for their data systems, use cases (ways in which stakeholders want to use DWRi's data and/or software), and technical requirements. Additionally, since DWRi is engaging with USU (and potentially others in the future) for software development activities, it will be important to have a DWRi employee who is

participating in, guiding, and providing leadership (from DWRi's perspective) for software development activities. Having DWRi "product owners" would provide a necessary feedback loop between DWRi, stakeholders, and software developers to ensure that requirements, use cases, and user needs are met. DWRi may have a system for this already, but new and modernized software applications under development will require significant input and feedback at each step in the Technology Modernization Roadmap as indicated by the dotted lines/arrows in Figure 1. Under this task, we recommend that DWRi do the following:

1. **Identify software/system "products" to be developed or maintained by DWRi:** Identify specific software systems or products within DWRi's data management ecosystem. These products may be specific databases or software programs (or groups of programs) that provide specific functionality needed by DWRi for their operations.
2. **Assign a DWRi employee to be the "product owner" for each identified product:** Product owners would have responsibility for leading requirements identification, describing use cases, providing input on technology selection and development, overseeing the development and testing of the product, and ensuring that stakeholders and stakeholder requirements are identified, adequately documented, and ultimately met by the software product (Recommendation 63).
3. **Solicit user feedback and translate into actionable issues for software development:** At each major step in the Technology Modernization Roadmap and with each software product under development it will be important to solicit user/stakeholder feedback and then translate bugs, feature requests, and enhancements into actionable issues in GitHub that can be prioritized and acted upon by software developers. Feedback may be solicited via meetings, workshops, or other means appropriate for the user group.

TASK 5.2: Evaluate DWRi's website and reorganize functionality to better meet DWRi's internal needs and those of external stakeholders

GAPS ADDRESSED:

- **Gap 1:** Accumulated technical debt has made it difficult for DWRi to maintain or modify existing databases and software systems to meet operational needs.
- **Gap 27:** Elements of DWRi's website remain significantly dated.
- **Gap 28:** It is not clear what data products, services, and/or automated data retrieval tools are available for accessing and consuming DWRi datasets. DWRi also lacks clarity on who potential users are and what their goals are.
- **Gap 29:** There do not appear to be standardized and documented web service APIs or encoding formats for accessing and exporting data.

DWRi's website serves as a primary interface with both internal and external stakeholders. The Gap Analysis identified a significant need to update and enhance DWRi's website to better meet stakeholder needs (Recommendation 47). This will be especially important as DWRi modifies

existing software programs and implements new ones to help manage data. These applications may change the way DWRi presents data via their website, and there may be new opportunities for building additional dashboards or applications based on the core data systems DWRi will be modernizing. Some stakeholders will be unfamiliar at first with new software interfaces and will need to be introduced to those programs. Other stakeholders may have specific needs that are not currently being met by DWRi's website but could be met by adding additional functionality that interfaces with DWRi's underlying core data systems. Under this task we recommend that DWRi do the following:

1. **Identify stakeholders (data consumers) and their specific use cases:** DWRi's employees are their primary stakeholders, but many others outside of DWRi have use cases for using the data DWRi maintains. Stakeholders will have differing technical needs and capabilities, use cases, and desired ways of accessing DWRi's data (Recommendation 48).
2. **Refine and align DWRi's website and web applications with user needs:** Significant opportunity exists to not only modernize the look and feel of DWRi's website, but also to reorganize information on the website to feature data products, make products easier to find and use, and build on top of DWRi's core data systems to provide views of data that stakeholders need (Recommendation 49).
3. **Document, feature, and promote data products and APIs on DWRi's website:** DWRi maintains specific data products that include diversion measurement data, Water Use Program data, and eventually a fully developed Water Right Network dataset. These datasets have tremendous value not only as the operational datasets that DWRi maintains for distribution accounting activities, but also as a historical record of the water management activities within the state. Promoting these datasets with sufficient organization, access interfaces, and documentation via DWRi's website will make them more valuable to a greater number of potential users.

ADDITIONAL CONSIDERATIONS

The Gap Analysis identified additional gaps related to personnel, training, and policy that are not fully covered by previous sections of this Technology Modernization Roadmap. We introduce them briefly here for completeness.

Personnel

With respect to personnel, Gap 39 indicated a lack of dedicated data management roles and personnel for which it was recommended that DWRi consider hiring a professional data manager (Recommendation 29, 63) and considering data management system software "product ownership" to better facilitate development, operation, and maintenance of the software systems operated by DWRi. Recommendation 63 lays out key responsibilities for a prospective data manager, but, in short, they might include coordination of data management efforts across DWRi's teams, leadership in establishing data governance, and effective data quality control processes (i.e., the operational rules and procedures around DWRi's data), and assisting DWRi

employees with strategic data utilization (e.g., ensuring that the data are available in the right formats and via the right protocols to meet the needs of users within and outside DWRi). A data manager should also be involved in developing DWRi's information and data models and in ensuring that they are consistently used. Additionally, a data manager could serve as a product owner for data-management software to ensure that it meets required functional use cases. The data manager would not develop the software, but rather would help ensure that it works correctly and meets the needs of stakeholders that use DWRi's data products.

The data manager role would be large and may be difficult for one person to accomplish. As such, DWRi may need to consider hiring technicians to assist, particularly in the areas of data management (e.g., ensuring that data are reliably being collected and entered into DWRi's data systems) and quality control (e.g., monitoring potential data quality alerts generated by DWRi's software and reacting as needed). While partially covered by the tasks described in earlier sections of this document, there is still opportunity for DWRi to consider their current technical workforce to determine whether new personnel are needed in specific roles (see Recommendation 65) to meet operational needs moving forward—where operational needs include increasing volume of data and potentially additional types of data associated with programs like Utah's Agricultural Optimization Program (Recommendation 66).

Gap 40 pointed out that some River Commissioners lack important technical skills and that there is a need to increase the level of technical capability among River Commissioners through establishing minimum qualification and a training program (Recommendation 64). River Commissioners represent a very important component of DWRi's workforce, and the success of DWRi's modernized data systems depends on use of these systems by River Commissioners and other DWRi employees (i.e., Distribution Engineers, Accounting Modeler(s), and others). As new software systems come online, DWRi will need to train employees to use the modernized systems they are building.

DWRi's Relationship with Utah Division of Technology Services

An additional personnel issue pointed out in Gap 41 concerns whether DWRi's relationship with the State of Utah Division of Technology Services (DTS) could be or will need to be enhanced to better meet DWRi's technology needs. This will be important as DWRi moves away from legacy databases and software applications (that DTS has been assisting DWRi in operating) and implements modernized systems that will need to be similarly supported by DTS. Included in the tasks/subtasks above are required steps to determine appropriate deployment and hosting environments for DWRi's modernized data management software systems. Important considerations include whether DWRi wants to host databases/applications on-premises or in the cloud, whether DWRi wants to operate virtual machines versus employing serverless computing and/or software as a service, and whether there are technical/security requirements imposed at the state level that will impact available options. DWRi will need to make these decisions with DTS to ensure that the outcomes work for all parties. These decisions will determine software development and deployment strategies and represent an opportunity for DWRi to consider the operations and the computing infrastructure needed to operate their

modernized data systems (Recommendation 14) and the personnel they are supporting in specialized roles (Recommendation 65).

Data Governance and Policy

With regard to data governance and policy, the Gap Analysis recommended that DWRi work toward establishing data governance policies and practices that ensure data quality, security, and provenance (Recommendation 7). This is both a technical problem that can be partially solved through better software systems that enforce business rules and constraints and a policy problem to ensure that data governance rules and procedures are established, understood, and executed. The recommended software and database development tasks above will largely address technical issues and enable the technical execution of data governance. However, DWRi will also have to decide how far-reaching their established data governance policies will extend.

The Gap Analysis suggested that DWRi consider the degree to which they want to use their statutory authority to require standardization of data submission, automated measurements/telemetry, and/or timely data reporting from partners and data providers (Recommendation 16). Requiring that data providers provide or submit data within a particular time frame, using a limited set of protocols, and using standardized formats would significantly simplify the software systems that DWRi has to operate and manage. However, this would pass technical requirements on to data providers—i.e., they may have to develop or adopt new software or scripts to push data to DWRi via DWRi's provided APIs or to make data available in a standardized format for DWRi to pull. This may work well for some data providers who have technical capabilities and can adapt their data reporting to DWRi's requirements but may be difficult for (or take significant time to implement for) data providers who currently lack these capabilities.

Finally, we reiterate that DWRi's operational data collection and management needs are not entirely unique. Recommendation 8 from the Gap Analysis encourages DWRi to consider their water data management systems in a holistic state context where multiple water-related agencies have similar data collection, management, and sharing requirements, and where sharing data and synthesis of datasets from different sources for important analyses would be easier if similar software, data interfaces, and data formats were used. Recommendation 8 of the Gap Analysis lays out potential benefits of a coordinated "Utah Water Information Center" for consideration by DWRi as they move forward with the tasks suggested in this Technology Modernization Roadmap and as they consider the holistic context of water management in the state of Utah and future coordination with Utah's water-related agencies.

4. Summary

This Technology Modernization Roadmap is organized into a set of four major task areas that are sequenced to enable DWRi to proceed with modernization of their core data systems related to measurement data (e.g., observations of flow recorded at stream diversions), Water Use Program data (water diversion data primarily from public water suppliers), and water distribution accounting (the procedures used by DWRi to ensure that water is distributed to water users by priority of legal water rights). A fifth major task area deals with stakeholder engagement and is intended to be a part of each of the other four major task areas.

The major task areas and suggested tasks and subtasks in this document were designed to address the gaps and recommendations identified and described in the Gap Analysis, and the specific gaps and recommendations are indicated throughout the text of this document where they are addressed. While some of the sequencing described in this document is based on our assessment of the priority for each identified gap/recommendation, many of the gaps and recommendations from the Gap Analysis ranked highly when considering the prioritization criteria we used. Thus, we determined that priority need alone was not enough to determine an order of operations. Many of the tasks and subtasks suggested in this document are interdependent and must be sequenced in a particular order. The main purpose of this document is to lay out a roadmap and suggested order of operations for DWRi to use for modernizing their data systems.

We anticipate that the extensive software development activities described in this document will take DWRi multiple years to implement. The recommended tasks in this document represent a comprehensive overhaul of major software and data management systems on which DWRi depends for its daily operational work. This will be a challenge not only for development, testing, and implementation of the software and systems, but also integration with DWRi's existing workflows and practices. However, doing so will enable DWRi to implement and operate reliable systems for managing Utah's operational water management data into the future and will support DWRi in meeting their mandate to "...administer the measurement, appropriation, apportionment, and distribution of the State's valuable water resources."

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